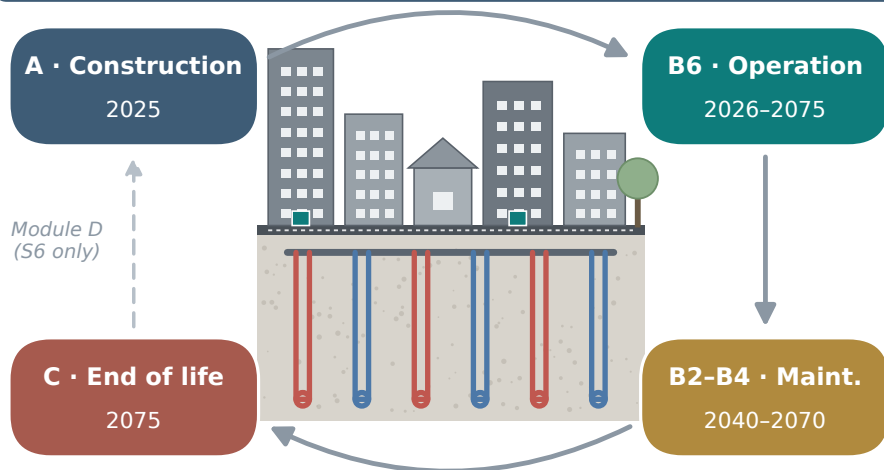


When does a geothermal energy network pay back its carbon?

Time-explicit, prospective and dynamic-climate LCA of networked geothermal (GEN) vs a gas/oil reference (REF)

Two systems deliver one identical service; every life-cycle stage is dated on a calendar (EN 15978)



GEN: shared borefield + heat pumps · REF: gas/oil furnaces + electric AC

identical delivered service: 89.375 GWh_{th} · 37 buildings · 50 years

One inventory, three methods: the result depends on time

2025 ————— 2075

every exchange dated and linked to a decarbonizing grid (premise · SSP2)

Conventional static — 44%
frozen 2025 grid

Time-explicit static — 28%
future grids, static factors

Dynamic climate — 42%
future grids + emission timing

GEN vs REF, GHG emissions (GWP100, kt CO₂-eq)

GEN repays its construction carbon by ≈ 2036 and emits 42% less by 2075

